Write a DeckGL layer for pie charts

DeckGL doesn't natively support pie charts. However, you can create a custom layer to display pie charts on the map by extending one of the existing layers or writing a new one from scratch.

Here is an example of how you might create a custom DeckGL layer for pie charts by extending the `CompositeLayer`. This example uses the `SolidPolygonLayer` to draw each slice of the pie. The data should include the longitude, latitude, and an array of values for each pie chart.

Below is a basic example of how you can create such a layer:

javascript

```javascript
import {CompositeLayer, SolidPolygonLayer} from '@deck.gl/layers';
import {getRadiusInPixel, toRadians} from './utils';  // You need to implem

class PieChartLayer extends CompositeLayer {
  renderLayers() {
    const {data, radius, getColor} = this.props;

    const pieSlices = data.reduce((acc, d) => {
      const [longitude, latitude] = d.position;
      const values = d.values;
      const total = values.reduce((sum, val) => sum + val, 0);
      let startAngle = 0;

      values.forEach((value, i) => {
        const percentage = value / total;
        const angle = percentage * 360;
        const endAngle = startAngle + angle;
        const radiusInPixels = getRadiusInPixel(radius, latitude);
```

```
          const slice = {
            ...d,
            startAngle: toRadians(startAngle),
            endAngle: toRadians(endAngle),
            radius: radiusInPixels,
            color: getColor(d, i)
          };

          acc.push(slice);
          startAngle = endAngle;
        });
```

ChatGPT 5.2

```
      }, []);

      return new SolidPolygonLayer(this.getSubLayerProps({
        id: 'pie-slices',
        data: pieSlices,
        getPolygon: d => getArc(d.startAngle, d.endAngle, d.radius),
        getFillColor: d => d.color,
        getElevation: d => 1000  // You can customize the elevation here
      }));
    }
  }

  // You need to implement this function to convert the start and end angles
  // and the radius into a polygon that represents a slice of the pie.
  function getArc(startAngle, endAngle, radius) {
    // TODO: Implement this function
  }

  PieChartLayer.layerName = 'PieChartLayer';
```

Note: This code is not complete and is meant to be a starting point. You would need to fill in the details and handle edge cases according to your specific requirements. You also need to implement the `getRadiusInPixel` and `toRadians` utility functions, and the `getArc` function to generate polygons representing each slice of the pie based on the start and end angles and the radius of the pie.

You would also need to provide the `getColor` prop to determine the color of each slice, and the `radius` prop to determine the size of the pie charts. The `data` prop should be an array of objects, each containing a `position` property with the longitude and latitude, and a `values` property with the array of values for the pie chart.

thank you

You're welcome! If you have any more questions or need further assistance with DeckGL or any other topic, feel free to ask. Happy coding!